

Project Proposal

Project Title

AI-Enhanced Community-Based Missing Children Reporting and Tracking System

1. Executive Summary

The safety and protection of children remain a major concern in many communities. In South Africa, cases of missing children occur frequently and require quick responses from both law enforcement agencies and community members. However, information about missing children is often shared through informal communication channels such as social media, messaging groups, or word of mouth. These methods can result in delayed communication and reduce the chances of locating missing children quickly.

According to missing persons organizations, approximately 1,697 children are reported missing each year in South Africa, which means that a child is reported missing approximately every five hours. Although about 77% of missing children are eventually found, delays in sharing information can significantly affect response time and recovery efforts.

This project proposes the development of an AI-Enhanced Community-Based Missing Children Reporting and Tracking System, a centralized web-based platform that enables community members to report missing children, share information, and receive alerts about active cases. The system will allow users to submit missing children reports, upload identifying images, and access updated case information.

In addition, the system will incorporate Artificial Intelligence (AI), Data Science, and Analytics techniques to improve the effectiveness of the platform. AI-based similarity analysis will help detect duplicate reports, while data analytics will identify patterns and trends in missing children reports across different locations and time periods.

The project will be developed over a six-month period using modern web technologies such as Java Spring Boot, PostgreSQL, HTML, CSS, JavaScript, and Python-based data analysis tools. The goal of the system is to improve communication, increase community awareness, and support faster responses when a child goes missing.

2. Background

2.1 History

Missing children's incidents have been a persistent concern for communities and law enforcement agencies worldwide. In South Africa, many families depend on police reports and social media platforms to spread information when a child goes missing.

While these approaches can raise awareness, they are often unstructured and decentralized, which can delay the distribution of critical information.

Over time, technology has played an important role in improving communication and information sharing in many sectors. Web-based systems and digital platforms have been widely used to manage information, improve collaboration, and provide real-time access to data.

Despite these technological advancements, many communities still lack a centralized digital platform specifically designed to report and track missing children's cases. As a result, important information may be scattered across different platforms, making it difficult for communities to coordinate efforts and respond quickly.

This project aims to address this challenge by developing a centralized reporting and tracking system that improves communication, enhances community awareness, and supports faster response during missing children's incidents.

2.2 Requirements

2.2.1 Business Problem

When a child goes missing, time is a critical factor in ensuring the child's safe recovery. However, many communities currently lack a structured and centralized system for reporting missing children and sharing information effectively.

Information about missing children is commonly shared through:

- Police reports
- Social media platforms
- Community messaging groups
- Word-of-mouth communication

Although these methods help spread awareness, they are often fragmented and difficult to manage. Important details may not reach community members quickly enough, and there is usually no centralized platform to store, organize, or analyse the information.

As a result, communities may experience delays in response, which can reduce the chances of locating the child quickly.

2.2.2 Business Opportunity

The increasing availability of digital technologies provides an opportunity to develop a centralized web-based system that allows communities to report and share information about missing children more efficiently.

A structured reporting platform could allow community members to:

- Report missing children quickly
- Upload identifying information and images
- Access updated reports within their community
- Receive alerts when new cases are reported

Additionally, the use of Artificial Intelligence and Data Analytics can enhance the system by detecting duplicate reports and identifying trends in reported cases.

Such a system can improve communication, encourage community participation, and provide a reliable platform for managing missing children reports.

This opportunity must be seized within the six-month development window (March to August 2026), as the growing reliance on digital platforms by South African communities presents a time-sensitive gap in the market for a structured, centralized missing children reporting solution. Delayed implementation means continued dependence on fragmented communication channels, further reducing response times and recovery rates.

2.3 Expected Solution

The proposed solution is the development of an AI-Enhanced Community-Based Missing Children Reporting and Tracking System.

Goal of the Solution

The main goal of the system is to create a centralized digital platform that enables community members to report missing children, share information quickly, and improve awareness during missing children's incidents.

The system will also incorporate Artificial Intelligence and Data Analytics techniques to improve data management and support more effective information analysis.

Scope of the Solution

In Scope

- Web-based reporting platform
- User registration and login
- Submission of missing children reports
- Optional image upload for identification
- Viewing and searching reports
- Community alert notifications
- AI-based duplicate report detection using similarity analysis
- Basic analytics to detect duplicate reports
- Secure storage of information

Out of Scope

- Integration with police databases
- Facial recognition technology
- GPS tracking of children
- Mobile application development

Approach

The project will follow the Software Development Life Cycle (SDLC) methodology, which includes the following phases:

1. Planning and requirements gathering
2. System design
3. System development
4. Testing and debugging
5. Implementation and evaluation

This structured approach will ensure that the system is developed efficiently and meets the project objectives.

3. Proposal

3.1 Vision and Goals

Vision Statement

By developing a centralized community-based digital platform for reporting missing children, this project aims to improve community awareness and response time by at least 30% within a six-month development period, enabling faster sharing of information and improving efforts to locate missing children.

Goals

Goal 1

To design and develop a functional web-based reporting system within six months that allows registered users to submit missing children reports containing information such as name, age, location, physical description, and images.

Goal 2

To implement secure authentication and data storage mechanisms by Month 3 (1 May 2026) that protect sensitive information, ensuring that no unauthenticated user can access or submit missing children reports, and that all passwords are encrypted using industry-standard hashing.

Goal 3

To implement AI-based duplicate detection that identifies at least 80% of duplicate reports, and an analytics dashboard displaying reporting trends by location and time period, fully integrated and testable by Month 4 (1 June 2026).

3.2 Project Deliverables

Title	Description	Notes
System Requirements Document	Documentation describing system requirements and design	Produced during planning phase
Web-Based Reporting Platform	Functional system for submitting missing children reports	Core system component
Database System	Secure storage of user information and missing children reports	Implemented using PostgreSQL
AI-Based Duplicate Detection Module	Uses similarity analysis to identify duplicate reports	Improves data accuracy
Analytics Dashboard	Displays statistics and reporting trends	Supports data analysis
Alert Notification Feature	Notifies users of new missing children reports	Improves community awareness

Final Project Documentation	Complete documentation including system design, testing, and evaluation	Submitted at project completion
-----------------------------	---	---------------------------------

3.3 Timeframe

This project must be delivered within six months to meet the academic requirements of ISJ107V and to address the urgent community need for a centralized missing child reporting system. Delivering after the allotted period would mean the solution fails to contribute to the required academic outcomes, and more critically, prolongs the absence of a tool that could directly improve response times when children go missing in South African communities. With approximately 1,697 children reported missing per year — one every five hours — every month without a better reporting system represents a real community risk.

Month 1 (16 March) – Research and Problem Analysis

Conduct research on missing children’s cases and identify system requirements.

Month 2 (6 April) – System Design

Design the system architecture, database structure, and user interface.

Month 3 (01 May) – Backend Development

Develop server-side functionality and implement authentication and report submission features.

Month 4 (01 June) – Frontend Development and Integration

Develop the user interface and connect the frontend with the backend system.

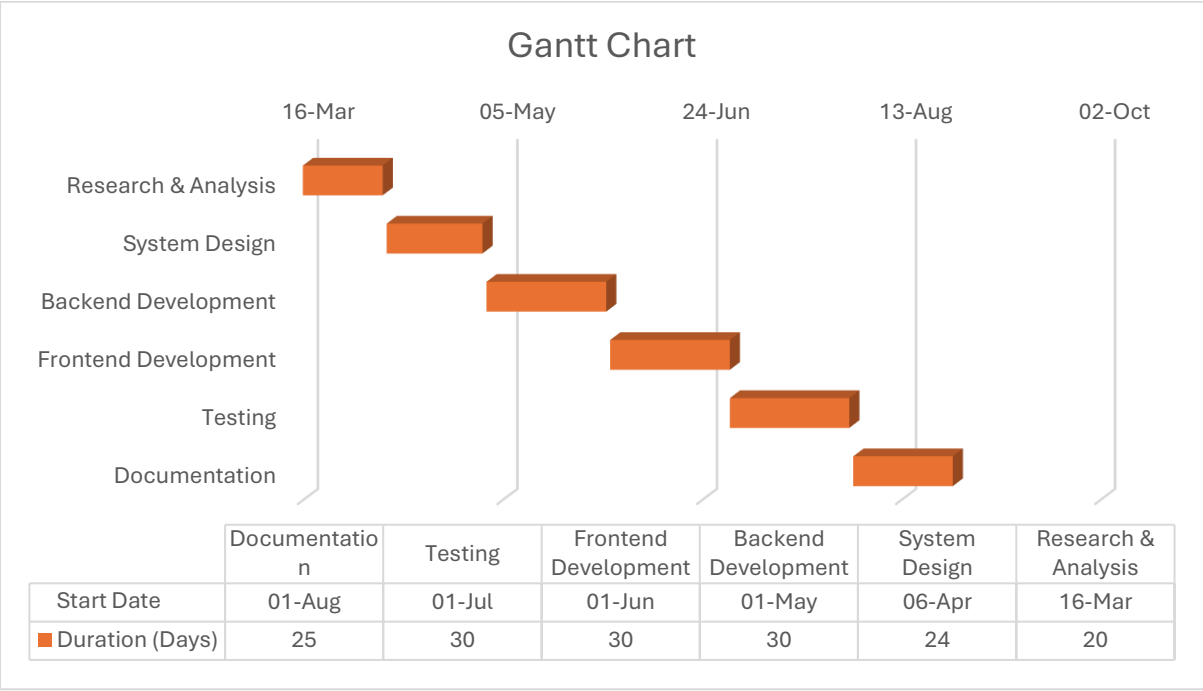
Month 5 (01 July) – Testing and Evaluation

Perform system testing and improve system performance.

Month 6 (01 August) – Documentation and Final Presentation

Prepare final documentation and demonstrate the completed system.

Task	Start Date	Duration (Days)
Research & Analysis	16-Mar	20
System Design	06-Apr	24
Backend Development	01-May	30
Frontend Development & Integration	01-Jun	30
Testing & Evaluation	01-Jul	30
Documentation & Final Presentation	01-Aug	25



3.4 Resources

Type	Quantity	Notes
Software Developer	1	Responsible for system design and development
IntelliJ IDEA	1	Used for backend development
Visual Studio Code	1	Used for frontend development
PostgreSQL Database	1	Used for storing reports and user data
Python Data Analysis Tools	1	Used for implementing AI and analytics features
Git Version Control	1	Used for code management
Laptop/Computer	1	Required for system development

3.5 Ethical and Security Considerations

Because the system will store sensitive information related to missing children, appropriate security and ethical considerations must be applied.

The system will implement secure authentication, controlled access to reports, and protection of personal data to ensure responsible handling of sensitive information. Only registered users will be allowed to submit or access detailed reports.

These measures will help ensure that the system protects user privacy and maintains responsible use of information.

3.6 Success Criteria

The success of the project will be measured using the following criteria:

1. The system is successfully developed and delivered within the six-month project timeframe.
2. The platform allows users to register, log in, and submit missing children reports successfully.
3. The system securely stores and manages user data and uploaded images.
4. The system provides alert notifications and search functionality for missing children reports.
5. The AI-based duplicate detection and analytics features successfully identify reporting patterns and potential duplicate cases.
6. The final documentation and system demonstration clearly show that the project objectives have been achieved.

4. Bibliography

Missing Children & South Africa Context:

Briefly.co.za. 2025. *2,000 of South Africa's children reported missing since 2022, 742 still missing*. [Online]. Available at: <https://briefly.co.za/south-africa/208918-2000-south-africas-children-reported-missing-2022-mzansi-saddened-742-missing/> [Accessed: March 2026].

Independent Online (IOL). 2025. *Failed systems, delayed responses a major let-down for SA's missing children*. [Online]. Available at: <https://www.iol.co.za/sundayindependent/dispatch/failed-systems-delayed-responses-a-major-let-down-for-sas-missing-children> [Accessed: March 2026].

Independent Online (IOL). 2025. *Exploring the impact of poverty and instability on missing children in South Africa*. [Online]. Available at: <https://www.iol.co.za/business/investigation-exploring-the-impact-of-poverty-and-instability-on-missing-children-in-south-africa-b6fae9cf> [Accessed: March 2026].

International Centre for Missing and Exploited Children (ICMEC). 2024. *Missing Children Statistics*. [Online]. Available at: <https://www.icmec.org/missing-children-statistics/> [Accessed: March 2026].

South African Government News Agency. 2025. *Nearly 16,000 arrested as SAPS intensifies fight against crimes against children*. [Online]. Available at: <https://www.sanews.gov.za/south-africa/nearly-16-000-arrested-saps-intensifies-fight-against-crimes-against-children> [Accessed: March 2026].

AI, Duplicate Detection & Analytics:

Zeng, Z., Dikeledi, I. and Lv, Y. 2024. A bibliometric analysis of research trends for missing persons globally and in East Asia from 2000 to 2021. **PubMed Central**. [Online]. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10965029/> [Accessed: March 2026].

Ruiz Reyes, J., Congram, D., Sirbu, R. and Floridi, L. 2024. *Where are they? A review of statistical techniques and data analysis to support the search for missing persons*. SSRN. [Online]. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4704138 [Accessed: March 2026].

Technology Stack:

Spring.io. 2020. *Building REST services with Spring*. [Online]. Available at: <https://spring.io/guides/tutorials/rest/> [Accessed: March 2026].

Sommerville, I. 2016. *Software Engineering*. 10th Ed. Harlow: Pearson Education Limited.

Missing Children South Africa. 2024. *Missing Children Statistics*. [Online]. Available at: <https://www.missingchildren.co.za> [Accessed: March 2026].